

The Max-Cube Completely Positive Volume Radius Vanishes

Automated Math Discovery Candidate Counterexample

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Status. Candidate full negative answer to Remark 1.5(1) of Klep–Štrekelj–Zalar, likely valid pending expert review. Under the nontrivial normalized volume-radius interpretation of “does not get arbitrarily small,” the proposed max-cube largeness fails even for the larger positive-semidefinite cone.

Source Question

Klep, Štrekelj, and Zalar compare compact affine sections of the copositive and completely positive cones and obtain a dimension-free lower bound for the ratio of their volume radii [1]. In Remark 1.5(1), on pages 4–5, they ask whether the analogous largeness persists inside the max-norm unit ball $B_{n,\max}$:

Remark 1.5. (1) There are two possible interpretations of the statements of Theorem 1.4. One is that the section $\widetilde{\text{CP}}_n$ is relatively large, while the other is that $\widetilde{\text{COP}}_n$ is relatively small. We believe that the first interpretation is more likely, as we now explain. Let $B_{n,\max}$ be the unit ball in the max norm $\|[a_{ij}]_{i,j=1}^n\|_{\max} = \max_{i,j} |a_{ij}|$ on $\mathbb{S}_n$. (This is not a proper matrix norm, since it is not submultiplicative.) Pushforwarding the Lebesgue measure on $\mathbb{R}^{\dim \mathbb{S}_n}$ to $\mathbb{S}_n$, equipped with the usual Frobenius inner product, it is clear that $\text{Vol}(\text{NN}_n \cap B_{n,\max}) = (\frac{1}{2})^{\dim \mathbb{S}_n} \cdot \text{Vol}(B_{n,\max})$ and hence $\text{Vol}(\text{COP}_n \cap B_{n,\max}) = c^{\dim \mathbb{S}_n} \cdot \text{Vol}(B_{n,\max})$ for some $c \in [\frac{1}{2}, 1]$. It follows that $\text{COP}_n \cap B_{n,\max}$ occupies a large portion of $B_{n,\max}$. Theorem 1.4 suggests that also the portion of $B_{n,\max}$, which is occupied by $\text{CP}_n \cap B_{n,\max}$, does not get arbitrarily small as $n \rightarrow \infty$, which is more surprising. However, this does not follow directly from Theorem 1.4, but remains as an open problem for future work. Comparing sizes of cones depends on the inner product and the compact

A RANDOM COPOSITIVE MATRIX IS CP WITH POSITIVE PROBABILITY

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sections chosen, and it is conceivable that in $B_{n,\max}$ the difference between the cones becomes visible (see also §2.2 below).

The displayed source passage writes the relevant volume as $c^{\dim \mathbb{S}_n} \text{Vol}(B_{n,\max})$ and is embedded in a discussion of volume radii. We therefore formalize the substantive question by asking whether the base c , rather than the raw probability $c^{\dim \mathbb{S}_n}$, stays bounded away from zero. (For raw probability, $\mathcal{CP}_n \subseteq \mathcal{NN}_n$ already gives an immediate negative answer.)

Definitions and Answer

Let $\mathbb{S}_n$ be the real vector space of symmetric $n \times n$ matrices, $d_n = n(n+1)/2$, and

$$B_{n,\max} = \{A = [a_{ij}] \in \mathbb{S}_n : \max_{i,j} |a_{ij}| \leq 1\}.$$

For a measurable cone $K \subseteq \mathbb{S}_n$, define its normalized max-cube volume radius by

$$\rho_n(K) = \left(\frac{\text{Vol}(K \cap B_{n,\max})}{\text{Vol}(B_{n,\max})} \right)^{1/d_n}.$$

This ratio is independent of whether Lebesgue measure is expressed in the independent upper-triangular coordinates or in orthonormal Frobenius coordinates.

Write $\mathcal{CP}_n$ for the completely positive cone and $\mathcal{PSD}_n$ for the positive-semidefinite cone. Let

$$v_k = \text{Vol}\{x \in \mathbb{R}^k : \|x\|_2 \leq 1\} = \frac{\pi^{k/2}}{\Gamma(k/2 + 1)}.$$

Theorem 1 (Negative answer in the max cube). *For every $n \geq 2$,*

$$\rho_n(\mathcal{CP}_n) \leq \rho_n(\mathcal{PSD}_n) \leq \frac{1}{2} \left[\left(\frac{2}{n+1} \right)^{n-1} \prod_{k=1}^{n-1} v_k \right]^{1/d_n}. \quad (1)$$

Consequently,

$$\rho_n(\mathcal{CP}_n) = O(n^{-1/2}) \longrightarrow 0.$$

Thus the max-norm ball makes the difference visible: completely positive matrices do not occupy a dimension-free normalized volume radius there.

Proof Intuition

The key point is that the max cube interacts very rigidly with positive semidefiniteness. Once the diagonal $(t_1, \dots, t_n)$ is fixed, every PSD matrix in the cube is uniquely a diagonally scaled correlation matrix $D^{1/2}CD^{1/2}$. The Jacobian contributes only $\prod_i t_i^{(n-1)/2}$. It remains to estimate the volume of the correlation ellipsope.

Build a correlation matrix one row and column at a time. Above a leading block C , the new correlation vector lies in the Schur-complement ellipsoid $x^T C^{-1}x \leq 1$, whose volume is $v_k \sqrt{\det C} \leq v_k$. Iteration gives a product of Euclidean ball volumes. That product has log volume $-(n^2/4) \log n + O(n^2)$, which becomes a factor of order $n^{-1/2}$ after taking the d_n -th root.

Proof

Let

$$\mathcal{E}_n = \{C \in \mathcal{PSD}_n : \text{diag}(C) = (1, \dots, 1)\}$$

be the correlation ellipsope, viewed in its $n(n-1)/2$ off-diagonal coordinates, and put $V_n = \text{Vol}(\mathcal{E}_n)$.

Lemma 1 (Ellipsope recursion). *For $n \geq 2$,*

$$V_n \leq \prod_{k=1}^{n-1} v_k.$$

Proof. Set $V_1 = 1$. Apart from a null boundary, a matrix in $\mathcal{E}_n$ has a positive-definite leading block and can be written

$$\begin{pmatrix} C & x \\ x^T & 1 \end{pmatrix}, \quad C \in \text{int}(\mathcal{E}_{n-1}).$$

By the Schur complement, the allowed fiber is $\{x : x^T C^{-1} x \leq 1\}$. Its volume is $v_{n-1} \sqrt{\det C}$. Hadamard's determinant inequality gives $0 \leq \det C \leq 1$, because C is PSD with unit diagonal. Fubini's theorem therefore yields

$$V_n = v_{n-1} \int_{\mathcal{E}_{n-1}} \sqrt{\det C} dC \leq v_{n-1} V_{n-1}.$$

Iteration proves the claim. $\square$

We next compute the PSD volume in the max cube. Use the independent entries a_{ii} and a_{ij} , $i < j$, as coordinates. The resulting volume ratios are the same as for the paper's Frobenius Lebesgue measure, since the latter merely multiplies every off-diagonal coordinate by $\sqrt{2}$ in both numerator and denominator.

If $A \in \mathcal{PSD}_n \cap B_{n,\max}$, then $t_i = a_{ii} \in [0, 1]$. Outside the null set on which some $t_i = 0$, put $D = \text{diag}(t_1, \dots, t_n)$ and

$$C = D^{-1/2} A D^{-1/2}.$$

Then $C \in \mathcal{E}_n$. Conversely, every $t \in (0, 1]^n$ and $C \in \mathcal{E}_n$ gives $A = D^{1/2} C D^{1/2} \in \mathcal{PSD}_n \cap B_{n,\max}$: indeed $|c_{ij}| \leq 1$ for a correlation matrix, so $|a_{ij}| \leq \sqrt{t_i t_j} \leq 1$.

For fixed t , the map on off-diagonal coordinates is $c_{ij} \mapsto a_{ij} = \sqrt{t_i t_j} c_{ij}$. Its Jacobian is

$$\prod_{i < j} \sqrt{t_i t_j} = \prod_{i=1}^n t_i^{(n-1)/2}.$$

Consequently,

$$\begin{aligned} \text{Vol}(\mathcal{PSD}_n \cap B_{n,\max}) &= V_n \prod_{i=1}^n \int_0^1 t_i^{(n-1)/2} dt_i \\ &= V_n \left(\frac{2}{n+1} \right)^n. \end{aligned} \tag{2}$$

In these coordinates $\text{Vol}(B_{n,\max}) = 2^{dn}$. Since $\mathcal{CP}_n \subseteq \mathcal{PSD}_n$, equation (2) and Lemma 1 give (1).

It remains only to read its asymptotics. The elementary ball-volume bound

$$v_k \leq \left(\frac{2\pi e}{k} \right)^{k/2}$$

implies

$$\begin{aligned} \log \prod_{k=1}^{n-1} v_k &\leq \frac{1}{2} \sum_{k=1}^{n-1} k (\log(2\pi e) - \log k) \\ &= -\frac{1}{4} n^2 \log n + O(n^2), \end{aligned}$$

where one uses $\sum_{k < n} k \log k = \frac{1}{2} n^2 \log n - \frac{1}{4} n^2 + O(n \log n)$. Because $d_n = \frac{1}{2} n^2 + O(n)$ and $n \log(2/(n+1)) = O(n \log n)$, taking logarithms in (1) gives

$$\log \rho_n(\mathcal{CP}_n) \leq -\frac{1}{2} \log n + O(1).$$

This proves Theorem 1. $\square$

Scope and Novelty Check

The theorem settles the max-cube question negatively in the normalized volume-radius sense. It does not compute the sharp constant or the exact asymptotic of $\rho_n(\mathcal{CP}_n)$; it only supplies the decisive vanishing upper bound. The separate source question about precise constants in the authors' special affine sections is unaffected.

Bounded searches through July 22, 2026 used arXiv:2305.16224, the exact source title, and combinations of “max norm,” “completely positive matrix,” “volume,” and “correlation ellipsope.” No later paper stating this answer or this reduction was found. The argument is short once the PSD/correlation change of variables is made, so novelty confidence is moderate and specialist review is recommended.

References

- [1] I. Klep, T. Štrelj, and A. Zalar, *A random copositive matrix is completely positive with positive probability*, arXiv:2305.16224 (2023; revised 2024).